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Writing Research Plans and Proposals

LS100 — Module 00A · Research Plans & Proposals

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This guide outlines the essential components of a clear, effective research plan or proposal. It assumes you have already built a research framework using *Research Methodologies: Converting the Problem to a Research Framework* (Guide 00) in this module; the question, operational definition, data model, and planned analysis you produced there are what these sections turn into a document. It breaks down each section — Title, Abstract, Introduction, Objectives & Questions, Methodology, Data Management, Ethics, Timeline, Expected Outcomes, and more — explaining their purposes and offering tips for coherence and rigor. The guide also provides critical-evaluation checkpoints and worksheet checklists for every section, enabling continuity from initial idea to final appendices. By following this guidance, students can develop a well-structured, feasible research plan that demonstrates scholarly grounding and practical significance.

1. Title

The title of your research plan should be concise yet descriptive. It is the first element readers see, and it creates an initial impression of your project. A strong title captures the essence of the study in a few informative words, allowing others to immediately grasp the topic. A good title should

1. Be specific, informative, scoped. Try keeping the title within 10 words.
2. Portray the central phenomenon, population or data source, and key variable(s) or approach.
3. Avoid cleverness that obscures meaning; prefer clarity.
4. If the study yields some novel findings, it is a good idea to put that on the title.

> **Critical Evaluation:**
>
> Test your title by asking: "Could someone guess the main question(s), method, and outcome from this line?"

2. 2. Abstract

The abstract is a compact overview of the entire plan, proposal or report, usually about 200–300 words. It functions as a mini proposal, summarizing your research

question, approach, and expected findings in accessible language. It should effectively distill the objectives, methodology, and significance of the project.

Although the abstract appears first, it is often written last — after you have developed the full plan — to ensure it accurately reflects the contents of the plan.

Critical Evaluation:

Does the abstract overreach (too many aims, vague methods) or under-specify (missing who/what/where/how)? Imagine a peer must decide in 60 seconds whether the study is coherent and feasible; write to that decision.

It should contain five elements:

1. **Problem context** - What gap or practical need does the work address?
2. **Research questions or hypotheses** (2–3 max) - The central questions or hypotheses guiding the study.
3. **Core methods** – The design, data source(s), analysis logic.
4. **Expected or potential outcomes** - Plausible results consistent with your logic (not promises, but what you anticipate finding).
5. **Significance** - Why it matters to scholarship or practice; the "so what?" of the study.

Good abstracts are tight but concrete: name variables, units of analysis, and comparisons (e.g., before/after; cohort A vs B). Avoid buzzwords unless you operationalize them (for example, instead of just saying "engagement," specify it as an index derived from "mean session duration, completion rate" – the variable you directly measured).

3. 3. Introduction & Background

The **introduction** sets the stage for your research, answering "What is the phenomenon, and why does it matter now?". Begin this section with the ***problem statement*** - a crisp, clearly articulated research problem or topic you will investigate and explain why it is important. Define the domain of your research (theory, experimental, practice, production, etc.).

> **Avoid asking tautological questions** - questions whose answer is guaranteed by the definitions involved, and no study could ever come out with different results. For example, "Does arm kinematics differ across different types of baseball or golf swings?" — logically, it must. No amount of study or data will find anything otherwise, and thus, not interesting. In contrast, questions like "Which specific running style minimizes joint stress on the patella during a marathon?" or "Does a 10-minute mindfulness break between morning lectures increase the sustained attention span of college students during afternoon seminars?" are genuinely investigative because their outcomes are not guaranteed by definition.

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Next, map the **knowledge landscape**: what is established, where uncertainty remains, and what debate or limitation your study targets. In crafting the background, demonstrate you are familiar with key scholarly work on the topic. Highlight how your work will build upon or depart from existing knowledge, and why those differences matter. By establishing the context and rationale, you show your project is grounded in scholarly understanding and addresses a meaningful question.

Provide enough background context for the reader to understand the broader field and where your project fits in. This often involves a brief literature review or discussion of prior studies to show what is already known and what gaps remain. Critically, guard against "literature dump." Each cited item must earn its place by informing design choices.

Next, clearly **articulate the gap**: identify a specific, **testable** missing piece in current knowledge or shortcomings in available tools that your study will address (e.g., prior work shows mean differences but not intra-subject variability; methods validated in adults but not adolescents; models trained in lab conditions, not field). State your **contribution type**—measurement, causal inference, replication, extension, synthesis, or tool building—and its expected payoff (better prediction, clearer mechanism, improved practice).

Your goal is to lead the reader toward your research vision: by the end of the introduction, they should understand the current state of knowledge, the gap or unresolved issue your study will address, and the significance of filling that gap.

Ensure **traceability** in your narrative: form a logical chain from background arrow.r research questions arrow.r methods. After this section, readers should smoothly transition into the Research Questions section.

> **LS100 Task:**
>
> Select 4–8 **anchor sources** (seminal studies, recent systematic reviews, or authoritative datasets) for the introduction of the research plan. For each, extract the actionable insight (claim, method, or limitation) that shapes your study.

4. 4. Research Objectives & Questions

After setting the context, clearly state your **research objectives** and central **research question(s)**. This section translates the broad aim of your study into specific goals and inquiries. In other words, objectives operationalize your vision into measurable aims; research questions (RQs) are the precise inquiries that satisfy those aims.

Research objectives break down the problem into actionable steps or outcomes you plan to achieve. Typically, a research plan will have a small number of key objectives (e.g. 2–4) that are **specific, clear, and achievable**, providing a roadmap for the

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project. For example, one objective could be to understand the relationship between A and B.

Begin by formulating one or more concise research questions that your project seeks to answer. For example, you might ask, "*What is the effect of A on B under conditions C versus condition D*?" – a question that pinpoints exactly what you will investigate. Each research question should be concrete and focused, delineating the problem you intend to solve or the hypothesis you will test.

For each RQ, specify the: **(i)** variables involved (predictors/outcomes), **(ii)** unit of analysis (frame, individual, trial, time in second/minute/ day), **(iii)** contrast or comparison (A vs B, pre/post, dose levels), and **(iv)** success criterion (what result would count as informative or supportive of your hypothesis). When appropriate, add a falsifiable **hypothesis**.

Finally, ensure **alignment** throughout your plan: objectives arrow.r RQs arrow.r measures arrow.r analyses arrow.r figures. Misalignment (e.g., posing an RQ about variability but reporting means) is the most common source of weak findings and should be avoided.

LS100 Task:

Keep **2 primary RQs** and **1 backup RQ** (activated if feasibility issues arise during the project). Ensure **construct validity** by tying each RQ to an operational definition (e.g., if you use an "engagement index," define it explicitly, such as engagement index = a function of {completion rate, dwell time, return frequency}). Map each RQ to one planned figure/table and one analysis method. Test for **Goldilocks scope**: too broad (unanswerable), too narrow (trivial), or just right - answerable with your data, and meaningful to the field.

5. 5. Methodology (Design arrow.r Materials arrow.r Procedures arrow.r Analysis)

The **methodology** section describes *how* you will carry out the research to achieve your objectives and answer your questions. It should be detailed enough that another researcher could in principle replicate the study, and it should demonstrate that your approach is well-reasoned and realistic.

Begin by outlining your research design and approach.

Design: State whether the study is experimental, quasi-experimental, observational, qualitative, or mixed-methods. Justify why this design is fit-for-purpose (causal estimation, measurement validation, theory building). Name threats to validity (confounding, selection, instrumentation) and how you will mitigate them (controls, matching, standardization). Include any ethical or compliance considerations if applicable (e.g. obtaining informed consent or data privacy measures for human subjects).

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Materials & Data Sources: Next, provide details on your data sources or participants, and the procedures for data collection. This includes describing the instruments and/ or tools you will use (such as surveys, questionnaires, interview guides, lab equipment, software, hardware), the target data or sample (including data units, inclusion/exclusion criteria), and expected sample size or data volume. Justify feasibility in practical terms (constrains of access, cost, time, etc.). Be explicit about why these sources and materials are appropriate and how they align with your research objectives — this shows that your choices will effectively address your questions. Also include any relevant logistical plans, such as how you will gain access to necessary data, facilities, or participant groups.

Procedures: Provide a reproducible, step-by-step workflow. This might include steps such as

data acquisition arrow.r preprocessing/cleaning arrow.r feature extraction/coding arrow.r quality checks. Where possible, externalize parameters in a "digital scaffold" (e.g., maintain a YAML/CSV file listing thresholds, labels, scoring rules) to make decisions explicit and auditable.

Analysis: Finally, explain how you will process and analyze the data (for example, using which statistical tests, coding, or computational models) and how this will help answer your research questions. Name primary analyses (statistical tests, models, coding schemes), assumptions (normality, independence), diagnostics (residual checks, intercoder reliability), and evaluation metrics (effect size, confidence intervals, F1 scores, Cohen's κ, etc.). It's helpful to tie each analysis back to its corresponding RQ and to a planned figure or table (e.g., "RQ1 will be addressed with a t-test comparing groups A and B, visualized in a bar chart of mean differences with 95% CIs."). Also, pre-specify decision rules: for instance, "If normality assumptions are violated, I will switch to a non-parametric test X; if $\kappa < 0.7$ in the coding, we will recalibrate the codebook and double-code additional samples."

Overall, the methodology must be **replicable**: a peer should be able to read this section and have a clear roadmap to recreate your study (with the help of your appendices for any detailed protocols).

6. 6. Data Management & Reproducibility

Reproducibility rests on how well you **organize, version, and document** your data and code. In this section, specify how you will manage your data and ensure that others (and future you) can follow your workflow.

Define a clear folder structure (e.g., /raw, /interim, /processed, /figures, /notebooks, /reports, /metadata). Adopt **consistent naming conventions** for files (e.g., YYYYMMDD_projectID_subjectID_session.ext) and maintain a README.md describing data flows and contents of each folder. Record **provenance** information: when and where data were collected, device/firmware versions, preprocessing parameters etc. Store parameters centrally (e.g., params.yaml) to ensure runs are

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traceable. Storing parameters in one place (say, a params.yaml file) can help keep track of analysis settings and ensure that runs are traceable.

Use version control (e.g., Git) for code and writing. For data, consider either storing immutable snapshots (with checksums for verification) or maintaining a manifest that references external data locations along with their hashes. Create a **reproducible computing environment** (for instance, provide a requirements.txt or environment.yml file) and document the commands or steps needed to run the analysis. For qualitative research, define versions of your coding scheme (codebook) and keep intercoder calibration logs.

Critically, specify **access control** measures: who can read/edit which folders, how sensitive data are anonymized, and where keys or consent forms are stored (e.g., in a separate secure location with restricted access). The goal is to make your project **hand-offable** – if you were unavailable, a peer should be able to pick up the project and continue with minimal confusion or loss of integrity.

7. 7. Ethics & Compliance

Ethical research safeguards participants, communities, and data subjects. Identify whether your project involves humans, animals, or sensitive data, and outline the ethical and compliance considerations.

Summarize any required oversight: for example, will you seek approval from an Institutional Review Board (IRB) or ethics committee? Determine if your project might qualify for exempt, expedited, or full review, and briefly explain why. For human subjects, detail your plans for informed consent (describe in plain language the study's purpose, any risks, the voluntary nature of participation, and the right to withdraw). Address privacy concerns: explain how you will anonymize or pseudonymize personal data and how long data will be retained. Also describe data security measures (such as encrypted storage and restricted access to data).

For observational studies or projects using scraped datasets, address terms of service, the distinction between public and private data, and any potential harms (e.g., the risk of re-identification or the use of data in ways that might stigmatize individuals or groups). If your project includes AI/ML components, include "**responsible AI**" considerations: check for dataset bias, plan for fairness audits, ensure transparency in how models make decisions, and avoid overstating what your model can do (especially in deployment).

Ethics sections should be **specific and practical**, not generic. State *exactly* how you will handle any particularly sensitive information (e.g., faces, names, GPS, or any sensitive/ protected attributes). Note where consent forms and decryption keys are stored (ideally separately from the data, with limited access), and who has role-based access. Close with a **risk-benefit** reflection: why anticipated benefits outweigh residual risks and what safeguards you will monitor throughout.

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8. 8. Timeline & Work Plan

A **timeline** (or work plan) translates your methods and objectives into a realistic plan with concrete schedule of tasks. It shows that you have thought through how to execute the project step by step within a given timeframe. One common approach is to break the project into phases or milestones (e.g., scoping & pilot, literature review, data collection, preprocessing/quality control, analysis & iteration, writing & dissemination) and estimate the duration of each.

You can present the timeline as a chart mapping tasks to specific weeks or months. For each phase, specify the key activities and goals. For example, a timeline might state:

- *Weeks 1–4: Conduct background literature review and refine research questions;*
- *Weeks 5–8: Collect data (through survey responses or experiments);*
- *Weeks 9–12: Analyze data and draft results.*

Under each broad phase, you can break down detailed weekly goals. You might also refer to the weekly timeline structure provided in the first presentation of LS100.

Each segment of the schedule should have clear **deliverables** or outputs (e.g., completion of an experiment, a draft chapter, "20 labeled sessions and $\kappa \geq 0.75$," "analysis scripts pass tests," "all planned figures generated") so that progress can be tracked.

Ensure that the sequence of tasks is logical (e.g., data analysis comes after data collection) and that you allocate sufficient time for each step. It's wise to include a buffer for unexpected delays, showing that the plan is feasible and not overly optimistic. The timeline should convince the reader that you have a **realistic plan** for the entire project from start to finish. It serves both as evidence of careful planning and as a practical guide to help you manage the research.

It's also a good idea to explicitly model potential risks in your timeline. For example, note points where you will evaluate progress and decide whether to invoke a backup RQ or an alternative method (decision points). Visualizing your plan in a simple Gantt chart or a table can clarify dependencies (such as needing ethics approval before recruitment, or having to complete data cleaning before analysis). Indicate where you have built in buffers or contingency plans (for instance, an extra week in case of slower data collection).

Feasibility is key: align your planned effort with real-world constraints (consider academic calendars, holidays, or limited lab access periods). Also reserve a final phase for polishing the report, formatting, and doing final reproducibility checks before submission. The timeline should ultimately persuade readers (and reassure yourself) that the project can be accomplished in the available time.

9. 9. Expected Outcomes & Impact

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In this section, outline the **expected outcomes** of your research and discuss its potential **impact**. Even though you haven't conducted the study yet, describe the results you anticipate or the products of the research (such as datasets, findings, new methods, or prototypes). Tying these expected outcomes back to your research questions shows how completing the study will fulfill the aims you set out.

Keep in mind that expected outcomes are **plausible projections**, not guarantees.

Be specific about the **analytic outputs** - for each RQ, you might describe the type of result you expect (e.g., "We anticipate a moderate improvement in accuracy ($\Delta \sim 10\%$) for model B over model A on dataset X" or "Survey group A is expected to report higher satisfaction levels than group B", or figures/tables tied to each RQ). These should be logical outcomes based on your framework or hypotheses, not just optimistic guesses. Also identify any **artifacts** you will produce, such as a cleaned dataset, analysis code, a new measurement instrument, or a prototype system. State how each outcome or artifact will address its corresponding RQ.

Impact has two dimensions: **Scholarly & Practical**.

- **Scholarly:** explain how results will fill a gap you identified, refine a measure or existing theories, replicate/contrast prior work, or introduce a reusable method.
- **Practical**: discuss who might use the findings and how. For instance, could teachers, clinicians, policymakers, or industry professionals use your findings?

Make sure to address the "*So what?*" question – why the outcomes matter and who might use or benefit from them.

It's also important to acknowledge **limitations** in advance. Note what the study might *not* be able to do (e.g., limited generalizability due to sample size or scope, potential measurement error, etc.) and how you will interpret the results in light of those limitations. Additionally, consider a **"null finding" scenario**: if you end up with no significant effects or opposite results, what then? Explain what knowledge would still be gained even if the main hypothesis isn't supported (such as discovering a boundary condition for a theory, or providing guidance that a certain intervention doesn't work under the tested conditions). By including this "**null path**," you show that the research has value regardless of outcome and that you have thought through all possibilities.

Highlighting the expected outcomes and impact demonstrates the value of the project. Essentially, you are answering what difference the research could make if successful – both within your academic field (advancing knowledge) and in a broader context (real-world application).

10. 10. References

A professional research plan must include a **References** section. Here you list all sources cited in your proposal, formatted in a consistent academic style (e.g., APA, MLA, Chicago, IEEE, etc.).

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Apply your chosen format uniformly to all types of sources (articles, books, datasets, software, etc.). A well-curated reference list gives credit to previous work, shows the foundation of existing knowledge on which your project builds, demonstrates your command of the field, and enables verification. Ensure every in-text citation corresponds to a full reference entry and vice versa.

Prioritize **high-quality sources**: peer-reviewed articles, authoritative books, major datasets with DOIs, and official documentation for software and tools. For fast-evolving domains, include recent reviews or position papers to show awareness of the latest developments.

Cite **for a reason**: every reference should support a claim, justify a method, or define a key concept. Avoid quoting at length; it's usually better to paraphrase and perhaps include a page number if a specific detail is critical. For software and data, include identifying details such as version number and access date.

It's helpful to maintain a **bibliographic log** while reading: for each source, record the claim, method, limitations, and how it affects your design. This prevents padding and helps you write the introduction efficiently.

You may use reference management software to organize your citations. Tools like Zotero (free, open-source), Mendeley (free, requires a university login for an account), and EndNote (subscription-based) can help you keep track of sources and automatically format citations in your chosen style.

11. 11. Appendices & Deliverables

Appendices are materials included with your proposal that provide supporting details without cluttering the main narrative. They are generally part of the supplementary materials. Examples of what you might put in appendices include: **A. Instruments & Protocols** (e.g., surveys, interview guides, device settings), **B. Data Schema** (a detailed variable dictionary, file formats, units of measurement), **C. Figure Specification Sheets** (for each planned figure, a description of what will be shown: axes, groupings, statistical annotations, etc.), **D. Consent Materials & Scripts**, **E. Parameter Files** (any configuration files like thresholds or label lists), and **F. Computing Environment Specs** (such as a requirements.txt or environment.yml listing the libraries and versions used).

Each appendix should be clearly labeled (e.g., "Appendix A: Survey Questionnaire") and referenced at the appropriate point in your proposal narrative (for example, "See Appendix A for the full questionnaire" or "…see Appendix C for details on the planned figures"). Keep appendices well-organized and only as long as needed; brevity with precision is the goal. Think of appendices as a toolkit for replication – if someone wanted to repeat your study, the appendices should provide every detail that isn't in the main text.

Deliverables for submission will typically include: the research plan itself (as a PDF or document), a README file explaining the project structure or how to run

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accompanying materials, and a minimal reproducibility package (for instance, a parameters file, a sample dataset if permissible, and code or notebook stubs). These materials together show that you are prepared not just to propose research, but to carry it out in a transparent and organized way.

Ensure the proposal itself is understandable on its own. The main text should be understandable without constantly flipping to appendices – appendices are there to provide depth for the reader who wants to know more. Together, thorough references and well-organized appendices convey professionalism and attention to detail in your research plan.

12. 12. Flexibility & Risk Mitigation

A research plan is a **blueprint**, but real-world research may require flexibility. Unexpected challenges and new findings are part of the research journey, so be prepared to adapt your plan if needed. A strong plan accounts for changes — for instance, by outlining alternative methods if the primary approach fails, or by including contingency time for delays. Treat your research plan as a living document: follow it closely to stay on track, but remain agile and open to refining your objectives or methods as you learn more. Striking this balance between structured planning and adaptive problem-solving will help ensure that your project stays on course and maintains rigor, even if circumstances change.

Remember, **plans guide; data decide**. Build optionality into your design: have a backup RQ, alternative measures, or a non-parametric analysis path ready in case assumptions are violated.

Identify the top risks to your project (e.g., difficulty recruiting participants, equipment failures, sensor drift, missing data issues) and set up early "detectors" – for example, pilot tests, quality control plots, or preliminary power analyses, QC plots etc. – that can alert you to problems while there's still time to pivot. If changes become necessary, document them transparently (keep a change log with what you altered and why) to maintain the credibility of your work.

Cultivate **interpretive humility**: be clear about what is exploratory versus confirmatory in your analysis, and label results accordingly. If you must deviate from the original plan, explain why the new path is still aligned with your core objectives and how you will ensure the validity of the study despite the changes. Showing that you can adapt thoughtfully and responsibly is far better than rigidly sticking to a failing approach.

13. Final assembly checklist (submission-ready)

Before submitting your research plan, use this final checklist to ensure all elements are clear, aligned, and complete:

- Title – Is it clear, specific, and appropriately scoped?

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- Abstract – Does it succinctly cover the context, RQs, methods, expected outcomes, and significance?
- Introduction – Does it tie the literature review to a concrete gap and clearly state your contribution?
- Research Questions – Are RQs aligned with the measures and analyses you'll use, and have you defined what success looks like for each?
- Methodology – Is the methodology detailed and replicable? (Check that parameters are documented externally, and that you've set up assumption checks and fallback plans.)
- Data Management – Have you ensured provenance, version control, and access control for your data and code?
- Ethics – Are ethical considerations addressed specifically (consent procedures, privacy protections, data security, fairness in algorithms, etc.)?
- Timeline – Is the schedule feasible, with logical sequencing, clear milestones, buffers for risks, and decision points for pivots?
- Expected Outcomes – Are the expected outcomes and impacts stated realistically (neither over-claimed nor vague), and do you discuss limitations and the value of potential null results?
- References & Appendices – Are references formatted consistently and purposefully chosen? Are appendices complete, useful, and properly referenced in the text?

Each item above should be checked and if necessary, revised before you consider the proposal final. This final assembly checklist helps ensure your research plan is polished and professional, giving you the best chance of approval and providing a strong roadmap for your project.

14. Conclusion

In conclusion, this guide is designed to help you develop a **comprehensive and coherent research plan** from start to finish. By following the advice and checklists for each section, you ensure that your proposal tells a clear story: from identifying an important problem and gap in knowledge, through detailing a rigorous method and timeline to tackle it, to considering the outcomes, impact, and ethical dimensions of your work. The aim is not only to fulfill the requirements of an assignment or proposal review, but also to instill good research planning habits that will serve you in future projects.

Remember that a research plan is a living document – use this guide to plan thoroughly, but remain thoughtful and adaptable as your project evolves. In doing so, you will demonstrate both scholarly rigor and practical foresight. Good luck with your research, and we hope this guide empowers you to write a proposal that is both insightful and feasible, setting the stage for a successful project.

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